

Trends in the rainfall pattern over India

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ABSTRACT: New monthly, seasonal and annual rainfall time series of 36 meteorological subdivisions of India were constructed using the monthly rainfall data for the period 1901–2003 of fixed network of 1476 rain gauge stations. In the new network, on an average, there is one rain gauge station for every 3402 Sq km area. The new rainfall series is temporally as well as spatially homogenous. Linear trend analysis was carried out to examine the long-term trends in rainfall over different subdivisions and monthly contribution of each of the monsoon months to annual rainfall. During the south-west monsoon season, three subdivisions viz. Jharkhand, Chattisgarh, Kerala showed significant decreasing trend and eight subdivisions viz. Gangetic WB, West UP, Jammu and Kashmir, Konkan and Goa, Madhya Maharashtra subdivision, Rayalseema, Coastal AP and North Interior Karnataka showed significant increasing trends. It has been found that the contribution of June, July and September rainfall to annual rainfall is decreasing for few subdivisions while contribution of August rainfall is increasing in few other subdivisions. EOF analysis is also done to know the spatial distribution of rainfall. The all India Monthly, seasonal and annual rainfall series constructed based on the 1476 stations are also reported. Copyright © 2007 Royal Meteorological Society

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1. Introduction

In the context of climate change, it is pertinent to ascertain whether the characteristics of Indian summer monsoon are also changing. The Indian summer monsoon (June–September) rainfall is very crucial for the economic development, disaster management, hydrological planning of the country. In spite of growth in the service sector, economy of India is still dependent on agriculture. Crop failure, drought and more extreme cases like famine due to weak or deficient monsoons becomes very critical to the country. Therefore, it is important to monitor closely the rainfall variation across the country on daily, weekly, monthly and seasonal time scale.

India Meteorological Department (IMD) after its establishment in 1875, took initiative to install more and more rain gauges around the country to capture the complex variation of rainfall across the country. Blanford (1886–1888) the first head of the IMD took personal effort to collect rainfall data from all the provinces and had brought out the detailed analysis of rainfall pattern over India. His work is well referred and is recognized as the most valuable information on rainfall distribution over India. Owing to these efforts of the IMD, a long-time series of rainfall data from a fairly uniform network of stations is available for research work. Many researchers used these precious data for

examining the spatial and temporal variation of rainfall across the country. Later, Mooley and Parthasarathy (1984), Parthasarathy *et al.* (1993), Parthasarathy *et al.* (1994), constructed all India rainfall series (denoted by Indian Institute of Tropical Meteorology Series (IITMS) here after) based on 306 uniformly distributed stations in the country. They have also used area weighted method to calculate all India rainfall using rainfall data of the 306 districts outside the hilly regions like Jammu and Kashmir, Himachal Pradesh, Hills of west Uttar Pradesh, Sikkim and Arunachal Pradesh, Bay Islands and Arabian Sea Island. Presently this time series is updated by the Indian Institute of Tropical Meteorology (IITM), Pune (www.tropmet.res.in) and this rainfall time series was extensively used by many researchers.

At present, there are more than 500 administrative districts in the country. It may not be ideal to represent the complex rainfall variation in the country using only 306 rain gauge stations as was done by Parthasarathy *et al.* (1993, 1994). Geographical area of all the administrative districts is more than 100 square km (except for Andaman and Nicobar Islands). Since rainfall over India has shown large spatial variability, more number of stations is essential to produce reliable rainfall data of administrative districts (WMO, 1983).

There were extensive studies on rainfall analysis on seasonal as well as annual time scales (Pramanik and Jagannathan, 1954; Parthasarathy and Dhar, 1978; Parthasarathy, 1984; Mooley and Parthasarathy, 1983, 1984; Parthasarathy *et al.*, 1993). All these studies discussed long-term trends of Indian monsoon rainfall for

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the country as a whole as well as for smaller regions. However most of these studies were based on the rainfall series constructed by Parthasarathy (1984). The monsoon rainfall is without any trend and mainly random in nature over a long period of time, particularly on the all India time scale (Mooley and Parthasarathy, 1984). Since rainfall is having high spatial variability, existence of long-term trend in smaller spatial scale was already reported by Koteswaram and Alvi (1969), Jagannathan and Parthasarathy (1973), Jagannathan and Bhalme (1973). In the subdivisional scale, existence of trends was noticed by Parthasarathy (1984) and Rupa Kumar *et al.* (1992). The study by Parthasarathy (1984) suggested that monsoon rainfall over two subdivisions viz. sub-Himalayan West Bengal and Sikkim and the Bihar Plains showed decreasing trends while for the four subdivisions viz. Punjab, Konkan and Goa, West Madhya Pradesh and Telangana showed increasing trends. However, Rupa Kumar *et al.* (1992) considered the linear trend analysis of each of the 306 stations of the IITMS and studied their spatial pattern. They have found significant increasing trend in monsoon seasonal rainfall along the west coast, north Andhra Pradesh and north-west India while significant decreasing trends over east Madhya Pradesh and adjoining areas, north-east India and parts of Gujarat and Kerala. A detailed discussion of other studies and their results are available in Pant and Rupa Kumar (1997).

The above mentioned studies used the data only up to the 1980s or so. Monsoon rainfall is known to have significant epochal variations (Pant and Rupa Kumar, 1997) and there was a significant change in the climate

pattern over India during the recent years. Therefore, it is ideal to extend the rainfall analysis with the recent data. Moreover, as mentioned above, these studies used the data of rainfall time series prepared using only 306 stations. Therefore, there was a scope to develop a homogenous rainfall time series with more number of stations and including the data of recent years. The scope of the present study is based on these two objectives, to develop a more homogenous long-time series rainfall data including the recent years and to examine the rainfall variations in more detail.

The development of a homogeneous (spatially as well as temporally) monthly rainfall data series was the first step in this study. From the vast data set archived at the National Data Centre, IMD, Pune, a network of 1476 rain gauge stations was selected that have only 10% or less missing years of data during the period 1901–2003. In the present network, we have covered all the area including the hilly region to have more uniformity. The rainfall analysis covered many aspects like linear trends, epochal variations, contribution of monthly rainfall to the annual total, teleconnection with the El Niño southern oscillation (ENSO) etc. As mentioned earlier, districts are the smallest administrative unit in India. There is a real need for district rainfall climatology for better hydrological and water management and also for agriculture and for any administrative purposes.

In Section 2, the details of the development of homogenous rainfall data series is discussed. The results are discussed in Section 3–9 and the conclusions are drawn in Section 10.

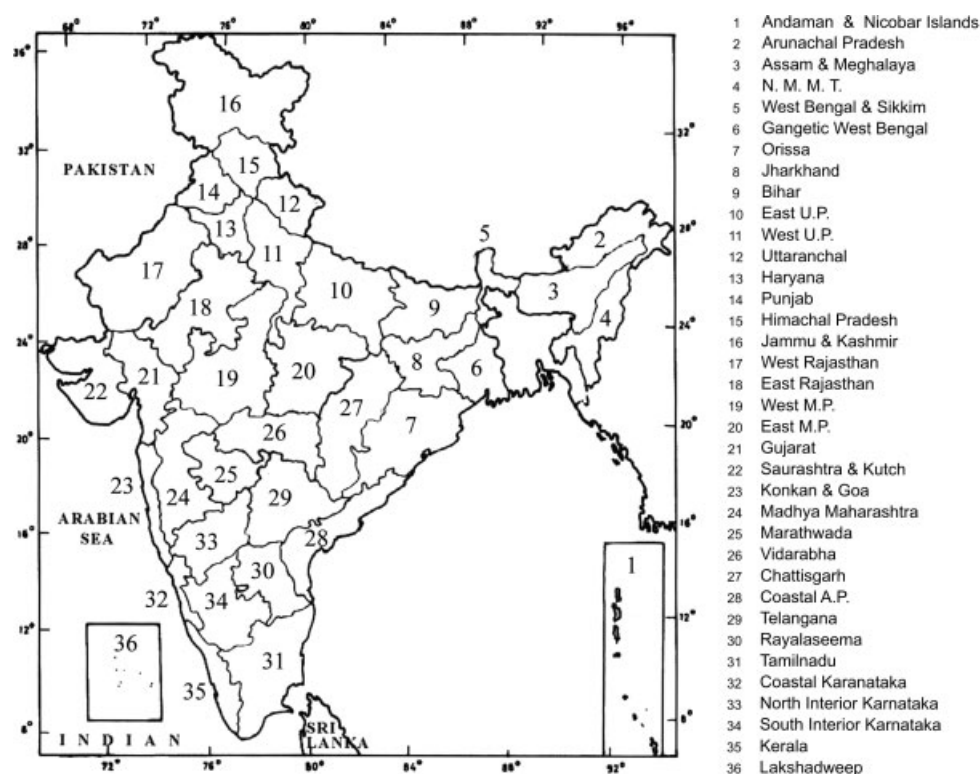


Figure 1. The 36 meteorological subdivisions of India.

Table I. Comparison of the number of stations, density and other statistical properties in each of the subdivisions used for IITMS and for the new one. The density i.e. the average area per station for each of the series is shown in the last two columns. Remaining 55 stations of the new network represent the other subdivisions which were not represented by IITMS.

Subdivision Name	Subdivision no.	No. of Station		Area (Sq Km)	Density/rain gauge	
		New network	IITM		IITM	New
Assam and Meghalaya	3	37	10	101012	10101.2	2730.1
N. M. M. T.	4	11	4	70447	17611.8	6404.3
West Bengal and Sikkim	5	6	5	28924	5784.8	4820.7
Gangetic West Bengal	6	13	11	66228	6020.7	5094.5
Orissa	7	17	13	155782	11983.2	9163.6
Jharkhand	8	46	6	79638	13273.0	1731.3
Bihar	9	71	11	94 238	8567.1	1327.3
East U.P.	10	101	26	146 509	5635.0	1450.6
West U.P.	11	107	19	96 782	5093.8	904.5
Haryana	13	37	12	45 821	3818.4	1238.4
Punjab	14	29	10	50 362	5036.2	1736.6
West Rajasthan	17	39	9	195 086	21676.2	5002.2
East Rajasthan	18	71	17	147 128	8654.6	2072.2
West M.P.	19	59	15	168 461	11230.7	2855.3
East M.P.	20	35	6	134 892	22482.0	3854.1
Gujarat	21	50	11	86 597	7872.5	1731.9
Saurashtra and Kutch	22	17	7	109 990	15712.9	6470.0
Konkan and Goa	23	32	5	34 095	6819.0	1065.5
Madhya Maharashtra	24	85	9	115 306	12811.8	1356.5
Marathwada	25	5	5	64 525	12905.0	12905.0
Vidarabha	26	42	8	97 537	12192.1	2322.3
Chattisgarh	27	28	22	139 488	6340.4	4981.7
Coastal A.P.	28	76	8	93 045	11630.6	1224.3
Telangana	29	10	9	114 726	12747.3	11472.6
Rayalaseema	30	42	4	69 043	17260.8	1643.9
Tamilnadu	31	182	15	130 549	8703.3	717.3
Coastal Karnataka	32	15	2	18 717	9358.5	1247.8
North interior Karnataka	33	33	6	79 895	13315.8	2421.1
South interior Karnataka	34	89	11	93 161	8469.2	1046.8
Kerala	35	36	10	38 864	3886.4	1079.6
Total		1421	306	2 866 848		

	Mean	Minimum	Maximum	Std.deviation	Range(Max-Min)
IITM	10566.47	3818.417	22482.00	4904.666	18663.58
New	3402.391	717.3022	12905.00	3159.548	12187.70

2. Development of a homogenous rainfall time series

To prepare a homogenous rainfall time series, we have selected 1476 rain gauge stations having maximum data availability during the period 1901–2003. Monthly rainfall data for these stations are available for at least 90% of the years. Missing data for individual station was filled up by the rainfall data of a neighboring rain gauge station. We have considered 458 districts for the present analysis. Each of these 458 districts of the country has two or more representing stations. First of all, the district monthly rainfall is calculated as the arithmetic average of rainfall data of all the stations in the district. Since there is a large spatial variability of

rainfall, it is necessary to apply the arithmetic mean for calculating areal rainfall up to district only. Rainfall data for the meteorological subdivisions were then calculated as the area weighted rainfall of the districts within the meteorological subdivisions. Figure 1 shows the location of the 36 meteorological subdivisions of the country. The average area of the meteorological subdivision is more than 91 000 sq km. It may be mentioned that the 306 stations used in IITMS cannot represent 458 districts. Table I gives the detailed comparison of the number of stations/density in each of the subdivisions used for the IITMS rainfall time series and for the new rainfall time series prepared by us. The density i.e. the average area per station for each of the series is shown in the last two columns. There are at least 14 subdivisions

where density of the IITMS time series is poor. On an average IITMS has one rain gauge station per every 10 566 sq km, whereas the new rainfall data time series has one station in each 3402 sq km area. Other statistical properties regarding the density of stations shown in Table I clearly indicate that the new series is spatially more homogeneous.

3. Rainfall over the country as a whole

All India monthly, seasonal and annual rainfall series were constructed based on the area weighted rainfall of all the 36 meteorological subdivisions of the country. The results are given in Table II. The mean, standard deviation and coefficient of variation are also given in the same table. The mean (calculated with the data of 1901–2003) rainfall of July is 286.5 millimetre is the highest and contributes 24.2% of annual rainfall (1182.8 millimetre). The August rainfall is slightly lower, which contributes 21.2% of the annual rainfall. The June and September rainfall are almost similar and they contribute 13.8 and 14.2% of the annual rainfall respectively. The mean south-west monsoon season (June–September) rainfall (877.2 millimetre) contributes to 74.2% of the annual rainfall (1182.8 millimetre). Contribution of pre-monsoon (March, April and May) rainfall and post-monsoon (October, November and December) rainfall to the annual rainfall is mostly the same (11%). Coefficient of variation is higher during the months of November, December, January and February. Figure 2 shows the comparison of the IITMS south-west monsoon season (June–September) rainfall series with the new rainfall time series. Even though the correlation coefficient between these two series is very large (0.97), there are many differences between the two series. The mean seasonal rainfall of IITMS series is 844.5 millimetre, whereas the mean value of this time series is 877.2 millimetre. The high mean value of the new time series is because of the consideration of all the 36 meteorological subdivisions, including the hilly regions. The standard deviation and coefficient of variability for the IITM series are 81.0 millimetre and 9.6% and the same for the new time series are 71.0 millimetre, 8.1% respectively. Coefficient of variation of the present time series is smaller compared to the IITM time series.

4. Epochal variations of Indian summer monsoon rainfall

It is well known that Indian summer monsoon rainfall displays multi-decadal variations in which there is a clustering of wet or dry anomalies (Pant and Rupa Kumar, 1997). To examine the epochs of above and below normal rainfall, a 31-year running mean of the Indian summer monsoon rainfall (ISMR) were calculated to isolate low frequency behaviour. These epochs of above and below normal rainfall are shown in Figure 3. Rainfall was above normal for nearly forty years from 1930s to 1960s. To understand the epochal behaviour of

rainfall series for different monsoon months, we have also calculated 31-year running means of each of the monsoon months (Figure not shown). It is seen that epochal behaviour of July and September rainfall is almost similar to that of the monsoon seasonal rainfall. In August, the above normal or positive phases started from the middle of 1950s and continued till to the end. Both June and August rainfall are in positive phase in the recent decades while July and September rainfall are in the negative phase. Figure 4 shows the decadal means of all India summer monsoon rainfall anomalies. The alternating sequence of multi-decadal periods having frequent droughts and flood years are clearly noticed in Figure 4. We can delineate the multi-decadal (1) 1901–1930 as dry period (2) 1931–1960 as wet period (3) 1961–1990 as dry period (4) 1991–2020 possibly wet period. Earlier studies by Pant and Rupa Kumar (1997) using the data series of Parthasarathy *et al.* (1994) also found similar results of 30 years of alternating sequences of dry and wet periods. **Table III shows the decadal mean, frequencies of drought and flood years. The deficient or excess monsoon years are defined for those years when monsoon rainfall percentage departures from the mean rainfall are less or more than the standard deviation (8.1% of mean). In the decade 1911–1920, there were four deficient and three excess years. During the dry period of 1901–1930, we had eight deficient years and three excess years. During the next three decades of wet period, we had three deficient years and five excess years. In the dry period of 1961–1990, there were seven deficient years and four excess years. During the period of 1901–2003, the number of deficient years (19) was more than the number of excess years (13).**

Figure 5 shows a similar picture for each of the four monsoon months. Except for the decade 1921–1930, the behaviour of July rainfall was almost similar to that of the monsoon seasonal rainfall. During the decade 1921–1930, in spite of high contribution from July, seasonal rainfall was negative because of higher negative contribution of June and August rainfall. Decadal variability is more in June where alternating sequence of wet and dry periods are seen on almost every decade. Coefficient of variability of July (12.3%) and August (12.0%) rainfall are also less compared to June (18.1%) and September (19.1%) rainfall.

5. Trends in all India monsoon rainfall

Figure 2 also shows the trend line drawn based on linear fit on the All India monsoon seasonal rainfall series as percentage departure from the long period average. The series was subjected to a 'low-pass filter' in order to suppress the high frequency oscillations. The weights used were nine point Gaussian probability curves (0.01, 0.05, 0.12, 0.20, 0.24, 0.20, 0.12, 0.05, 0.01). It is clearly seen that no linear trend exists in this series. We have also used linear regression technique and the 'Students *t*-test' for testing if there is any significance in the trend in

Table II. All India monthly, seasonal and annual rainfall in millimetre.

Year	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	J-F	MAM	J-S	O-D	ANNUAL
1901	34.1	40.2	29.6	41.9	59.1	129.4	252.7	268.6	137.3	59.5	37.1	14.0	74.3	130.6	788.0	110.6	1103.5
1902	11.4	12.4	28.6	48.1	57.3	123.8	285.6	209.8	201.1	69.9	29.3	27.3	23.8	134.0	820.3	126.5	1104.6
1903	18.7	14.0	35.8	28.8	66.3	131.4	298.9	269.3	195.3	116.1	39.3	22.6	32.7	130.9	894.9	178.0	1236.5
1904	17.5	15.6	38.5	38.5	77.2	169.2	271.9	216.4	141.6	73.7	13.4	19.7	33.1	154.2	799.1	106.8	1093.2
1905	24.9	26.3	45.1	39.5	62.3	112.1	263.5	211.3	175.6	60.2	12.9	14.2	51.2	146.9	762.5	87.3	1047.9
1906	23.0	49.2	39.1	25.5	45.6	185.3	290.7	252.2	182.5	55.7	19.2	29.8	72.2	110.2	910.7	104.7	1197.8
1907	15.8	48.6	46.7	66.8	41.8	160.9	236.3	310.9	104.0	31.8	24.5	16.6	64.4	155.3	812.1	72.9	1104.7
1908	22.2	23.0	21.5	38.2	52.5	135.7	327.0	308.5	158.8	46.8	8.9	12.6	45.2	112.2	930.0	68.3	1155.7
1909	25.7	21.4	19.3	69.4	59.9	208.1	314.9	229.0	165.9	45.2	12.6	31.4	47.1	148.6	917.9	89.2	1202.8
1910	16.2	15.2	22.5	35.8	50.6	213.2	251.3	285.5	191.6	111.8	36.1	9.5	31.4	108.9	941.6	157.4	1239.3
1911	41.1	11.1	52.9	32.4	52.5	196.8	174.0	214.6	181.3	71.0	43.8	14.8	52.2	137.8	766.7	129.6	1086.3
1912	23.4	23.7	27.5	43.9	49.7	115.3	329.3	262.2	128.8	61.3	50.3	8.5	47.1	121.1	835.6	120.1	1123.9
1913	8.9	41.7	31.0	33.4	77.1	218.8	278.5	198.3	117.9	69.6	18.7	25.1	50.6	141.5	813.5	113.4	1119.0
1914	6.4	32.8	33.2	50.6	72.5	166.9	348.6	239.8	198.2	52.6	22.3	23.2	39.2	156.3	953.5	98.1	1247.1
1915	22.1	42.1	51.8	42.8	62.0	161.8	232.9	225.8	175.8	93.8	47.6	11.2	64.2	156.6	796.3	152.6	1169.7
1916	5.8	22.3	17.8	36.0	57.6	215.0	269.8	302.6	197.4	140.4	45.5	5.8	28.1	111.4	984.8	191.7	1316.0
1917	9.6	35.7	29.7	43.2	80.0	221.3	267.4	287.3	277.6	157.1	27.4	13.4	45.3	152.9	1053.6	197.9	1449.7
1918	14.1	7.1	39.6	41.1	89.4	181.2	160.9	231.0	105.2	23.5	44.7	18.7	21.2	170.1	678.3	86.9	956.5
1919	50.6	24.4	27.6	34.9	60.0	185.7	294.3	288.6	152.6	77.1	50.4	22.5	75.0	122.5	921.2	150.0	1268.7
1920	24.9	23.6	49.4	36.8	57.5	151.4	294.2	177.9	122.3	47.0	26.5	6.2	48.5	143.7	745.8	79.7	1017.7
1921	39.9	10.8	20.0	41.1	40.1	172.5	274.6	259.9	193.5	69.3	16.8	19.3	50.7	101.2	900.5	105.4	1157.8
1922	29.4	12.5	18.4	32.7	47.4	184.4	304.1	214.4	200.6	56.5	55.2	16.5	41.9	98.5	903.5	128.2	1172.1
1923	26.0	42.2	29.0	32.4	55.3	98.6	321.1	272.2	167.9	63.0	17.7	18.3	68.2	116.7	859.8	99.0	1143.7
1924	21.3	25.7	20.7	35.1	59.7	121.3	315.0	249.4	232.9	63.3	54.6	18.7	47.0	115.5	918.6	136.6	1217.7
1925	14.6	13.4	20.9	42.3	86.2	199.2	297.7	232.2	123.4	72.2	42.4	18.4	28.0	149.4	852.5	133.0	1162.9
1926	29.4	13.2	59.0	42.9	59.6	95.1	301.8	326.6	205.5	54.9	11.8	11.7	42.6	161.5	929.0	78.4	1211.5
1927	14.3	34.2	27.6	33.5	54.5	164.5	333.6	251.9	152.6	62.8	56.9	13.7	48.5	115.6	902.6	133.4	1200.1
1928	23.2	42.5	25.6	39.8	48.6	158.8	291.3	216.4	139.1	115.3	23.4	29.1	65.7	114.0	805.6	167.8	1153.1
1929	28.0	22.4	18.4	50.4	55.0	179.0	292.6	240.9	122.9	95.8	19.6	39.7	50.4	123.8	835.4	155.1	1164.7
1930	23.6	22.2	27.1	47.2	59.8	172.2	289.1	196.7	173.5	93.1	47.7	12.6	45.8	134.1	831.5	153.4	1164.8
1931	13.8	33.2	22.2	33.6	56.4	114.5	294.7	305.5	186.3	121.7	41.2	23.8	47.0	112.2	901.0	186.7	1246.9
1932	10.0	25.1	26.7	32.6	72.1	125.0	326.7	227.8	173.8	68.9	55.7	17.2	35.1	131.4	853.3	141.8	1161.6
1933	17.8	32.9	31.0	48.7	98.8	206.9	275.7	301.9	211.1	95.9	21.3	19.1	50.7	178.5	995.6	136.3	1361.1
1934	24.8	11.4	22.2	36.5	41.5	197.5	273.5	290.6	164.4	62.2	29.5	14.7	36.2	100.2	926.0	106.4	1168.8
1935	30.0	21.3	22.7	46.9	36.2	141.9	312.3	228.1	178.1	57.1	17.3	12.9	51.3	105.8	860.4	87.3	1104.8

(continued overleaf)

Table II. (Continued).

Year	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	J-F	MAM	J-S	O-D	ANNUAL
1936	11.8	41.5	41.1	32.1	79.4	241.9	276.2	228.4	185.4	64.3	56.2	22.6	53.3	152.6	931.9	143.1	1280.9
1937	7.8	53.6	24.6	57.9	53.8	160.2	329.3	194.9	174.0	94.5	21.9	21.0	61.4	136.3	858.4	137.4	1193.5
1938	30.8	32.6	28.2	31.1	69.4	245.5	283.5	239.6	156.2	75.0	16.1	7.9	63.4	128.7	924.8	99.0	1215.9
1939	14.1	33.4	36.5	38.3	39.4	153.7	262.5	235.6	150.3	88.9	29.7	4.5	47.5	114.2	802.1	123.1	1086.9
1940	16.0	27.9	45.1	32.9	76.0	170.4	296.0	282.9	115.1	62.4	41.4	18.0	43.9	154.0	864.4	121.8	1184.1
1941	25.7	17.4	21.2	30.7	69.4	162.2	234.0	223.4	148.0	62.9	27.6	21.5	43.1	121.3	767.6	112.0	1044.0
1942	25.0	45.1	19.3	45.5	56.1	172.3	339.5	286.0	180.4	44.0	18.2	29.2	70.1	120.9	978.2	91.4	1260.6
1943	54.9	12.2	26.7	48.6	86.0	154.6	305.5	228.3	203.0	90.5	18.1	8.5	67.1	161.3	891.4	117.1	1236.9
1944	28.5	43.0	59.7	37.6	47.6	138.1	343.4	288.4	148.1	90.4	30.3	17.9	71.5	144.9	918.0	138.6	1273.0
1945	34.8	10.8	23.2	48.0	50.4	155.7	315.2	232.9	210.6	82.4	21.2	7.8	45.6	121.6	914.4	111.4	1193.0
1946	6.9	21.5	26.0	48.5	63.5	201.1	297.4	286.5	141.6	79.6	76.9	39.4	28.4	138.0	926.6	195.9	1288.9
1947	23.2	21.0	29.1	35.4	47.2	124.0	294.7	287.4	234.3	66.9	9.8	26.4	44.2	111.7	940.4	103.1	1199.4
1948	25.6	29.0	42.4	40.2	75.2	153.9	308.3	275.2	176.5	63.4	70.0	12.6	54.6	157.8	913.9	146.0	1272.3
1949	13.2	29.7	24.4	48.0	78.1	146.7	298.9	236.8	217.2	93.2	12.6	5.7	42.9	150.5	899.6	111.5	1204.5
1950	30.2	25.0	37.4	26.9	50.2	142.7	335.5	235.6	196.7	56.6	25.4	10.6	55.2	114.5	910.5	92.6	1172.8
1951	15.9	15.3	43.1	46.2	58.4	150.3	251.6	223.9	130.5	75.4	31.7	8.7	31.2	147.7	756.3	115.8	1051.0
1952	11.0	22.6	35.1	37.8	69.9	168.4	281.8	249.0	121.9	76.7	9.4	24.5	33.6	142.8	821.1	110.6	1108.1
1953	26.2	13.9	24.8	42.4	48.7	163.3	312.9	286.0	169.8	88.5	14.1	9.8	40.1	115.9	932.0	112.4	1200.4
1954	30.4	36.6	27.9	30.2	56.3	148.6	297.7	237.5	244.5	81.6	5.6	16.9	67.0	114.4	928.3	104.1	1213.8
1955	23.5	11.1	29.9	39.5	77.4	180.3	241.2	313.9	217.7	145.9	28.4	12.9	34.6	146.8	953.1	187.2	1321.7
1956	17.5	16.6	36.7	37.4	85.8	208.4	351.3	259.6	169.0	149.7	42.8	15.8	34.1	159.9	988.3	208.3	1390.6
1957	32.2	18.8	41.3	35.3	62.4	152.3	288.9	264.9	130.3	66.4	27.2	16.3	51.0	139.0	836.4	109.9	1136.3
1958	15.1	20.5	29.8	39.8	66.9	123.9	314.5	285.4	215.0	103.9	35.4	18.7	35.6	136.5	938.8	158.0	1268.9
1959	28.7	25.1	30.0	34.1	68.5	162.6	345.5	255.5	219.1	118.6	25.4	11.6	53.8	132.6	982.7	155.6	1324.7
1960	16.3	9.2	39.7	28.1	64.3	152.0	290.6	244.6	168.6	75.3	34.8	12.6	25.5	132.1	855.8	122.7	1136.1
1961	24.9	33.9	27.8	33.5	73.2	185.8	329.0	277.3	228.0	116.7	22.9	13.6	58.8	134.5	1020.1	153.2	1366.6
1962	14.9	24.1	26.3	47.3	63.6	123.5	272.9	257.6	207.0	80.8	19.9	31.5	39.0	137.2	861.0	132.2	1169.4
1963	13.5	16.2	38.7	45.6	58.7	148.8	254.4	294.9	163.2	93.0	25.9	19.1	29.7	143.0	861.3	138.0	1172.0
1964	13.2	19.0	27.9	38.0	53.6	152.3	320.6	273.2	198.4	68.5	26.4	14.8	32.2	119.5	944.5	109.7	1205.9
1965	13.3	22.9	34.2	41.6	50.9	121.0	279.4	210.1	145.5	45.3	18.7	25.4	36.2	126.7	756.0	89.4	1008.3
1966	15.7	23.8	24.8	34.9	59.8	169.1	253.1	229.5	151.7	62.1	49.4	18.4	39.5	119.5	803.4	129.9	1092.3
1967	12.9	14.1	55.0	33.1	48.5	146.3	296.4	266.3	176.8	52.0	14.8	46.1	27.0	136.6	885.8	112.9	1162.3
1968	24.3	21.4	30.5	37.0	45.6	146.5	302.4	214.8	144.3	73.2	23.4	14.6	45.7	113.1	808.0	111.2	1078.0
1969	12.7	16.4	26.9	42.5	66.0	131.8	305.2	260.6	179.6	63.5	35.5	17.5	29.1	135.4	877.2	116.5	1158.2
1970	23.2	27.4	33.3	37.4	66.1	195.5	248.8	300.8	203.3	75.2	20.0	10.0	50.6	136.8	948.4	105.2	1241.0

1971	17.4	23.1	26.3	49.0	73.7	203.3	260.3	260.3	159.2	95.5	16.1	18.1	40.5	149.0	883.1	129.7	1202.3
1972	12.4	25.3	24.8	37.4	60.4	137.1	226.3	234.1	138.9	70.8	31.3	23.8	37.7	122.6	736.4	125.9	1022.6
1973	15.0	19.4	27.5	31.9	61.3	148.5	284.5	293.4	182.7	106.2	17.8	19.3	34.4	120.7	909.1	143.3	1207.5
1974	11.7	14.4	25.0	35.8	66.2	126.0	271.6	240.0	155.8	101.1	14.4	11.9	26.1	127.0	793.4	127.4	1073.9
1975	17.3	21.3	32.5	31.5	54.5	176.1	295.3	284.8	224.1	109.8	23.8	9.8	38.6	118.5	980.3	143.4	1280.8
1976	12.7	22.2	32.3	39.6	50.4	157.7	294.1	292.1	150.2	39.0	53.5	13.0	34.9	122.3	894.1	105.5	1156.8
1977	17.9	18.4	27.9	58.3	81.3	184.3	305.7	254.9	152.2	88.7	61.8	15.1	36.3	167.5	897.1	165.6	1266.5
1978	14.8	28.6	40.7	32.9	64.1	187.2	291.4	277.9	160.4	62.7	44.5	24.6	43.4	137.7	916.9	131.8	1229.8
1979	20.0	34.7	34.1	26.9	56.9	151.1	244.4	235.1	146.4	63.5	71.8	16.4	54.7	117.9	777.0	151.7	1101.3
1980	15.0	21.5	31.7	38.8	50.6	212.2	290.4	263.3	144.8	56.3	26.0	23.4	36.5	121.1	910.7	105.7	1174.0
1981	26.9	20.3	45.3	36.1	60.9	158.9	303.6	239.3	193.5	53.8	29.1	16.8	47.2	142.3	895.3	99.7	1184.5
1982	25.9	25.1	41.1	50.0	63.3	139.9	242.7	274.5	135.5	60.3	43.9	15.0	51.0	154.4	792.6	119.2	1117.2
1983	18.2	23.5	38.4	49.7	63.8	150.3	279.8	294.4	224.4	85.1	14.0	24.1	41.7	151.9	948.9	123.2	1265.7
1984	20.4	31.8	31.0	42.6	56.3	164.3	282.1	260.7	149.7	69.1	18.2	16.9	52.2	129.9	856.8	104.2	1143.1
1985	23.0	14.4	30.8	36.6	54.2	153.0	270.9	238.7	157.6	115.0	20.9	22.5	37.4	121.6	820.2	158.4	1137.6
1986	17.8	34.5	28.5	48.5	53.9	174.1	265.7	235.2	142.0	72.0	43.9	25.9	52.3	130.9	817.0	141.8	1142.0
1987	17.7	20.4	29.0	39.4	64.0	130.1	237.6	237.1	152.0	88.4	45.2	21.3	38.1	132.4	756.8	154.9	1082.2
1988	11.0	24.9	43.8	45.6	68.3	159.9	353.4	285.2	214.0	57.1	17.9	17.0	35.9	157.7	1012.5	92.0	1298.1
1989	15.8	17.8	32.2	34.4	56.1	183.3	308.2	238.1	170.2	54.2	20.6	18.1	33.6	122.7	899.8	92.9	1149.0
1990	16.8	42.5	49.5	43.4	101.7	180.8	279.0	292.7	195.2	99.9	32.2	25.2	59.3	194.6	947.7	157.3	1358.9
1991	14.6	25.7	29.8	50.9	73.5	181.8	281.4	256.0	135.6	63.9	33.7	22.0	40.3	154.2	854.8	119.6	1168.9
1992	19.2	23.5	33.9	32.5	53.9	141.9	257.9	269.0	169.2	69.0	42.3	7.6	42.7	120.3	838.0	118.9	1119.9
1993	18.6	26.9	40.0	30.7	67.6	165.7	314.1	209.8	200.6	87.8	29.0	18.6	45.5	138.3	890.2	135.4	1209.4
1994	23.9	25.9	27.3	47.2	50.9	199.8	336.3	278.9	153.3	87.6	26.8	19.1	49.8	125.4	968.3	133.5	1277.0
1995	27.5	29.0	28.7	34.5	77.7	137.4	301.3	256.1	183.8	76.3	35.7	10.6	56.5	140.9	878.6	122.6	1198.6
1996	22.3	22.8	36.5	35.5	60.9	170.7	277.0	283.4	147.0	98.8	16.5	18.6	45.1	132.9	878.1	133.9	1190.0
1997	16.2	13.0	33.8	42.9	51.0	166.7	284.8	270.5	163.0	68.1	56.7	45.1	29.2	127.7	885.0	169.9	1211.8
1998	16.9	33.7	41.9	41.7	58.9	162.4	293.1	253.3	195.5	102.1	38.9	12.5	50.6	142.5	904.3	153.5	1250.9
1999	20.3	13.6	18.2	24.2	81.9	167.3	274.8	251.0	191.5	106.5	23.2	8.5	33.9	124.3	884.6	138.2	1181.0
2000	22.0	26.8	22.9	43.7	69.9	178.1	274.5	240.2	154.3	60.7	20.6	11.5	48.8	136.5	847.1	92.8	1125.2
2001	11.3	14.1	23.2	44.2	61.4	185.6	275.4	229.8	138.3	95.0	26.7	11.8	25.4	128.8	829.1	133.5	1116.8
2002	20.4	20.1	28.5	42.3	59.7	161.2	163.9	244.0	173.0	69.8	27.0	12.0	40.5	130.5	742.2	108.8	1022.0
2003	15.5	30.9	32.8	39.0	55.7	167.8	305.9	250.1	181.9	93.8	26.0	20.7	46.4	127.5	905.7	140.5	1220.1
Mean (1901–2003)	20.3	24.6	32.0	39.8	61.9	163.4	286.7	255.3	171.8	78.4	30.7	17.9	44.9	133.7	877.2	126.9	1182.8
S.D.	8.5	10.0	9.2	7.9	12.5	29.5	35.3	30.6	32.8	24.8	15.1	7.9	12.3	17.7	71.0	28.8	87.0
C.V.	41.8	40.4	28.8	19.9	20.2	18.1	12.3	12.0	19.1	31.6	49.3	44.0	27.3	13.2	8.1	22.7	7.4

Table III. Decadal mean (% departure from normal), frequency of drought and flood years.

DECADE	Decadal mean(%) departure from normal	No. of deficient years	No. of excess years
1901–1910	–2.2	3	0
1911–1920	–2.5	4	3
1921–1930	–0.4	1	0
1931–1940	1.7	1	1
1941–1950	3.3	1	1
1951–1960	2.5	1	3
1961–1970	–0.1	2	1
1971–1980	–0.8	3	1
1981–1990	–0.3	2	2
1991–2000	0.6	0	1
2001–2003	–5.9	1	0

each of the monsoon months. All India summer monsoon season rainfall as well the all India rainfall during the four monsoon months (June, July, August and September) does not show any significant trend.

6. Trends in subdivisional rainfall

Normand (1953) pointed out that ‘India as a whole is too large to be treated as a single unit. Some areas are negatively correlated with others, for example, the monsoon rainfall of Bengal and Assam with Bombay and central India’. For the country as a whole, the monsoon season rainfall and monthly rainfall for the

monsoon months do not show any significant trend. But there can be large variations in the regional scale. In order to study the secular variations of regional rainfall we have carried out the trend analysis for the monthly rainfall series of June, July, August and September and also for the monsoon season as a whole for all the 36 subdivisions.

The results are shown in Figure 6, which shows significant and remarkable variations on the regional scale. We have analysed July and August rainfall, which contributes to a major portion of the monsoon seasonal rainfall. We find in July, six subdivisions have shown decreasing trends and eight subdivisions have increasing trends. In August, four (ten) subdivisions have shown decreasing (increasing) trends in rainfall. We have considered all the cases of 99, 95 and 90% levels of statistical significance. June rainfall has shown increasing trend for the western and south-western parts of the country, whereas decreasing trends are observed for the central and eastern parts of the country. July rainfall has decreased for most parts of the central and peninsular India but increased significantly in the northeastern parts of the country. August rainfall has increased significantly (at 95% significance level) for the subdivisions Konkan and Goa, Marathwada, Madhya Maharashtra subdivision, Vidarbha, West Madhya Pradesh, Telangana and west Uttar Pradesh. The September rainfall has shown significant increasing trends (at 95% level of significance) in Gangetic West Bengal and decreasing trends (at 90% level of significance) for the subdivisions Marathwada, Vidarbha and Telangana.

Figure 7 shows the trends in south-west monsoon season rainfall (in millimetre in 100 year) for each of

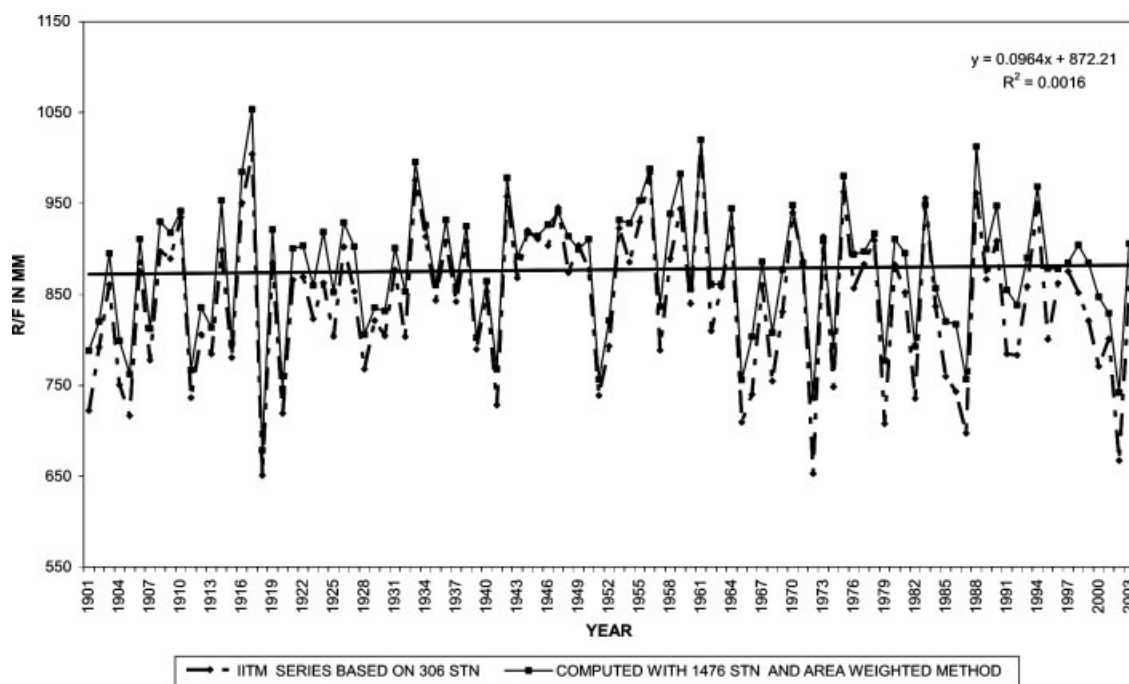


Figure 2. Comparison between IITM south-west monsoon seasonal rainfall series and the new series. Trend line drawn on the new series shows no trend in the seasonal rainfall.



Figure 3. The 31 year moving averages of all India south-west monsoon seasonal rainfall.

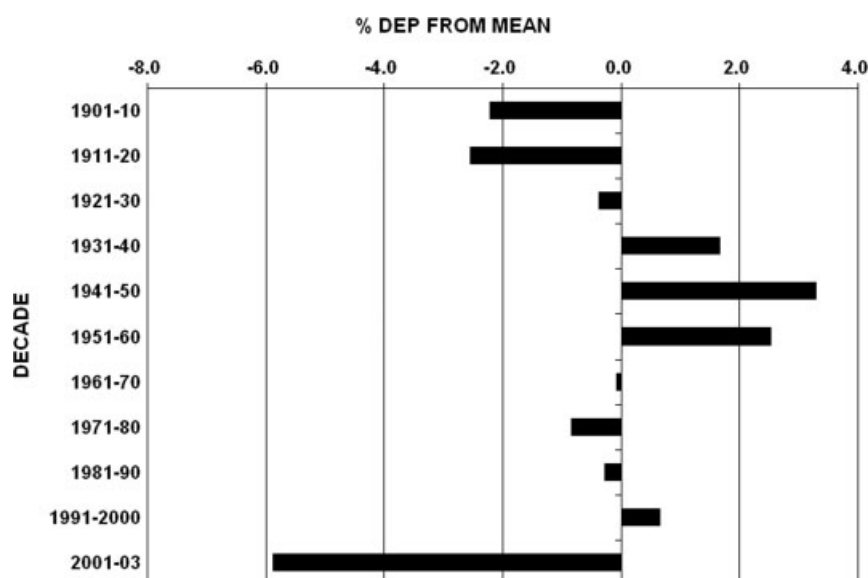


Figure 4. Decadal means of all India summer monsoon rainfall (% departure from mean).

the 36 subdivisions. Different levels of significance are shaded. During the season, three subdivisions viz. Jharkhand (95%), Chattisgarh (99%), Kerala (95%) showed significant decreasing trends and eight subdivisions viz. Gangetic WB (90%), West UP (90%), Jammu and Kashmir (90%), Konkan and Goa (95%), Madhya Maharashtra subdivision (90%), Rayalaseema (90%), Coastal AP (90%) and North interior Karnataka (90%) showed significant increasing trends.

In order to examine further, whether the contribution of each month's rainfall in the annual rainfall shows any significant trend, we have prepared a time series of contribution of rainfall for each month towards the annual total rainfall for each year. Trend analyses of per cent contribution of monthly rainfall to the annual rainfall were then carried out for each month and for all the 36 subdivisions. Results suggest that contribution of June and

August rainfall exhibited significant increasing trends, while contribution of July rainfall exhibited decreasing trends. Figure 8 shows very interesting results. June rainfall is gaining importance as its contribution to annual rainfall has increased in almost 19 subdivisions while it has decreased in the remaining 17 subdivisions. However a significant increase occurred in three subdivisions while a significant decrease occurred in four subdivisions. Contribution of July rainfall is decreasing in central and west peninsular India (significantly in South interior Karnataka (95%), East M.P.(90%), Vidarbha (90%), Madhya Maharashtra (90%), Marathwada (90%), Konkan and Goa (90%), and North interior Karnataka (90%)). Interestingly, contribution of August rainfall is increasing in all these subdivisions. Therefore, we see a major shift in the rainfall pattern spatially and temporally during the recent years.

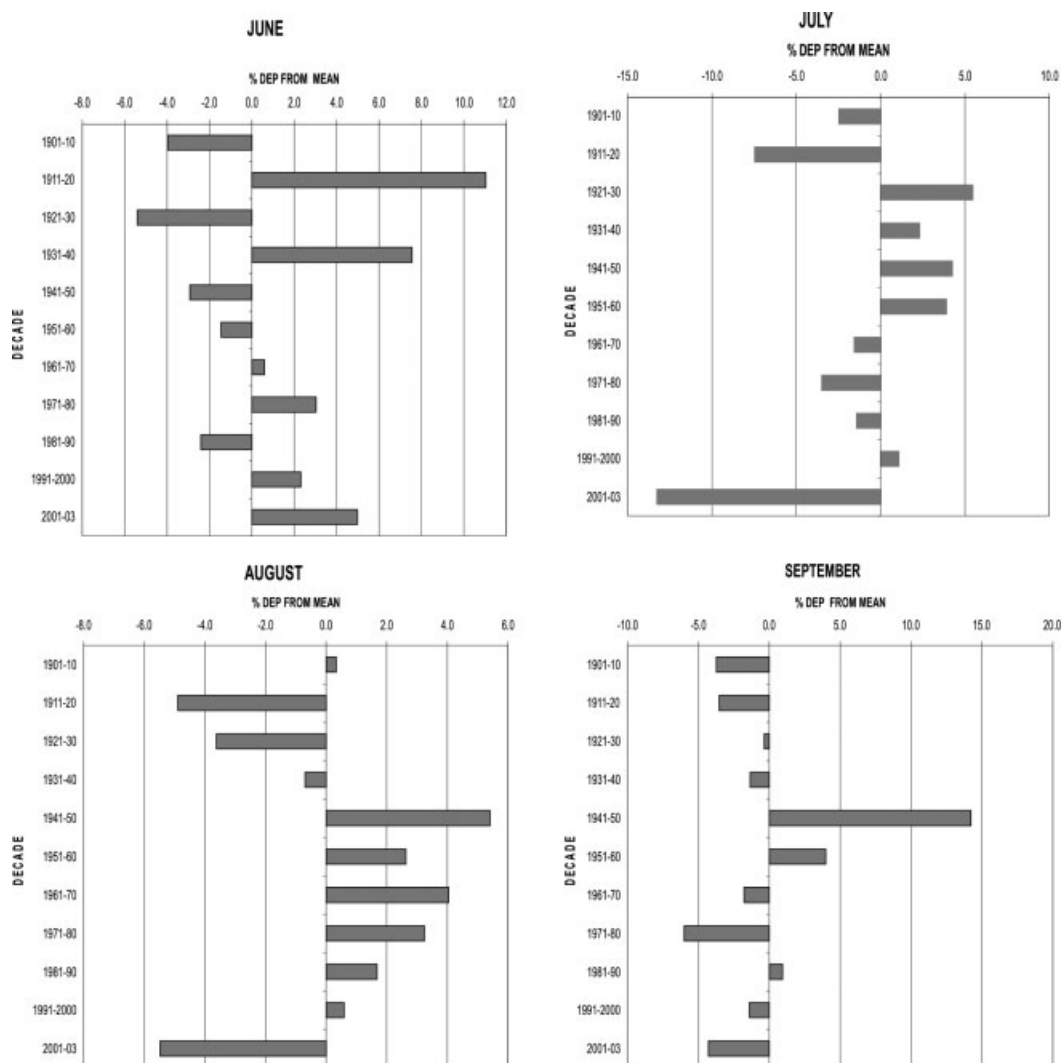


Figure 5. Decadal means of all India rainfall (% departure from mean) for the month of June, July, August and September.

7. Quantifying the importance of trend

A simple measure of the practical importance of trend in a time series is the fraction of original variance of the series accounted for by the fitted trend line, which can be computed by

$$R^2 = 1 - \frac{Var(e_t)}{Var(x_t)}$$

where $Var(x_t)$ is the variance of the original time series, and $Var(e_t)$ is the variance of the residuals from the trend line. The above equation measures the importance of the trend component in a time series to total variance and can range from 0 for no importance to 1 if the series is purely trend.

As stated earlier we have found that there is a significant (95% or more) decreasing trend in monsoon rainfall for the subdivisions Chattisgarh, Jharkhand and Kerala and an increasing trend for the subdivision Konkan and Goa.

The R^2 value for each of these four subdivisions are calculated and are found to be 0.09, 0.05, 0.05 and 0.04 respectively. In other words, the trend component

explains about 9, 5, 5 and 4% of the total variance of south-west monsoon rainfall of Chattisgarh, Jharkhand, Kerala and Konkan and Goa respectively.

8. Trends in subdivisional rainfall during other seasons

Though south-west monsoon is the major rain producing season over the country, other seasons have also significant contribution in some specific areas. Rainfall during the winter (January–February) and pre-monsoon (March–May) seasons are mostly predominant by western disturbances and convective activities over north and northwest India. Northeast monsoon is predominant over southern states during the October–December period. It may be mentioned that western disturbance is the term used in tropical countries like India, Pakistan, Nepal to describe an extratropical weather system that brings sudden rain and snow to the northwestern parts of the Indian subcontinent during winter and pre-monsoon months. This is a non-monsoonal precipitation pattern driven by the Westerlies. The moisture in these storms usually originates over the Mediterranean Sea and the Atlantic Ocean.

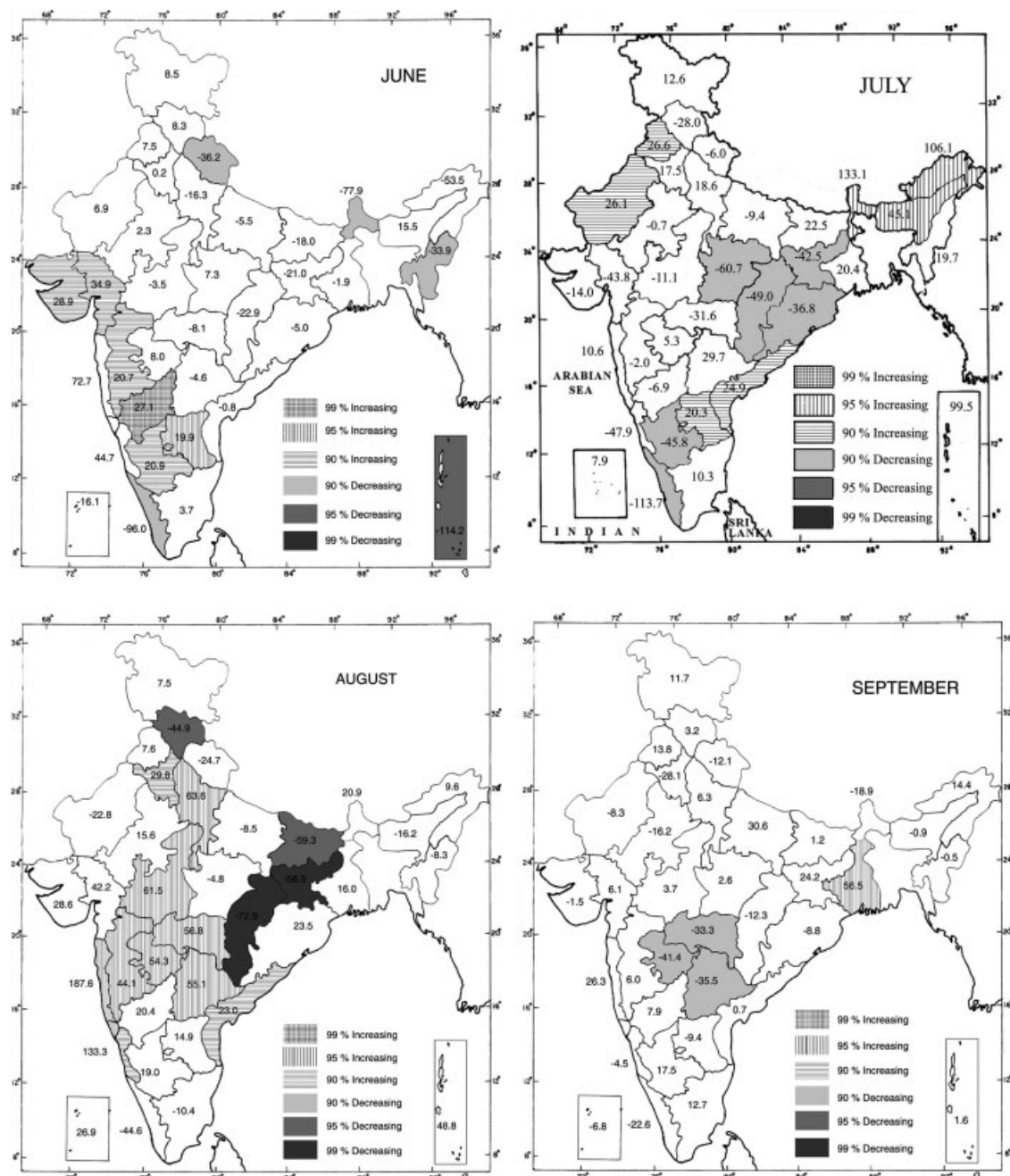


Figure 6. Increase/Decrease in rainfall in millimetre in 100 year for each of 36 subdivisions for the monsoon months. Different levels of significance are shaded.

Extratropical storms are a global, rather than a localized, phenomena with moisture usually carried in the upper atmosphere (unlike tropical storms where it is carried in the lower atmosphere). In the case of the subcontinent, moisture is sometimes shed as rain when the storm system encounters the Himalaya (Pisharoty and Desai, 1956).

Therefore, trends analysis was also carried out on subdivisional rainfall series for the winter season (January–February), pre-monsoon season (March–May), post-monsoon season (October–December) and also for the annual rainfall. Figure 9 shows the increase/decrease in millimetre in 100 years in each of 36 subdivisions for the winter, pre-monsoon, post-monsoon seasons and

annual period. Different levels of statistical significance are shaded. Rainfall has shown decreasing trend in almost all the subdivisions except for the subdivisions Himachal Pradesh, Jharkhand and Nagaland, Manipur, Mizoram and Tripura during the winter season. Rainfall for the subdivisions viz. east Uttar Pradesh, Bihar, east Madhya Pradesh where winter rainfall is mostly due to western disturbances is also decreasing significantly. Rainfall is decreasing significantly over 18 subdivisions of the country during the winter season. During the pre-monsoon season, rainfall is decreasing over most parts of the central India. This may indirectly suggests that the convective activity, which is the main cause for the rainfall activities during the pre-monsoon season is decreasing over

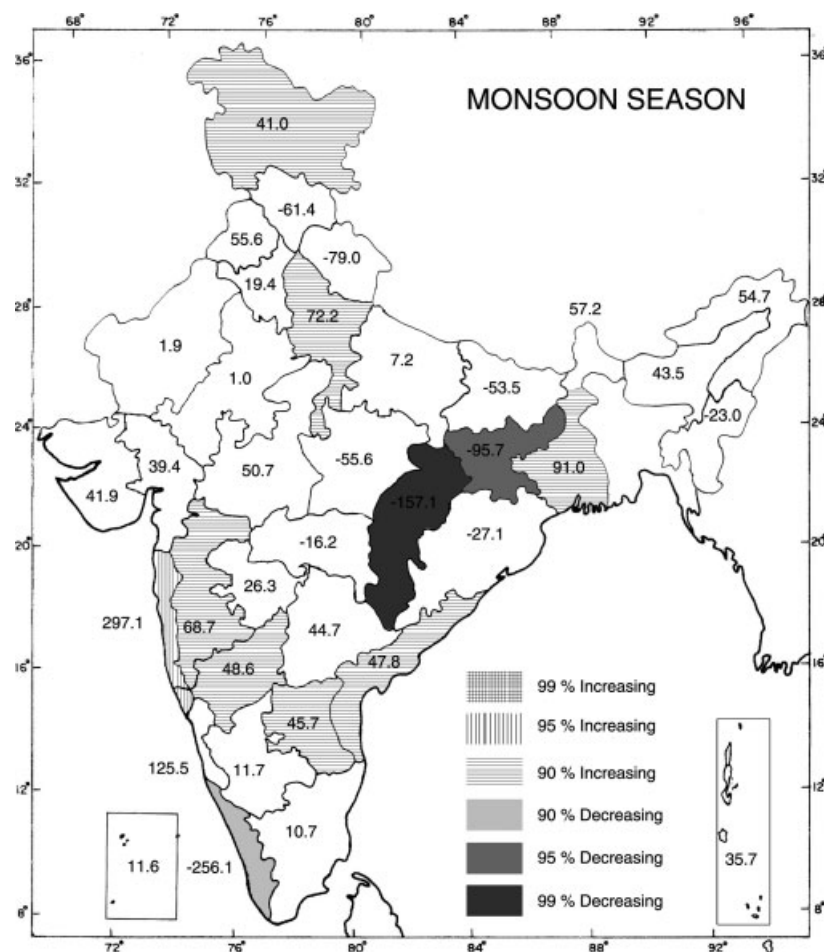


Figure 7. Increase/Decrease in rainfall in millimetre in 100 year for each of 36 subdivisions for the south-west monsoon season. Different levels of significance are shaded.

the central parts of the country. Rainfall is decreasing significantly for the six subdivisions viz. Gujarat Region, west M.P., east M.P., Vidarbha, Chattisgarh and Jharkhand. Also, according to the study by Das *et al.* (2002), the frequency of the western disturbances has decreased significantly. However during the post-monsoon season, rainfall has shown increasing trends in almost all the subdivisions except over nine subdivisions. Rainfall is increasing significantly over the subdivisions viz. Saurashtra and Kutch, Marathwada and Rayalseema. For the subdivisions Chattisgarh, Jharkhand and Kerala significant decrease in rainfall is even observed in the annual scale. Significant increasing trend is observed in the annual scale for the subdivisions Konkan and Goa, Madhya Maharashtra, North Interior Karnataka, Rayalseema, coastal Andhra Pradesh, Gangetic West Bengal, Assam and Meghalaya and Jammu and Kashmir.

9. EOF analysis of rainfall and its teleconnection with ENSO and other features

There is a large spatial variability in the Indian summer monsoon rainfall (Rasmusson and Carpenter, 1983; Pant and Rupa Kumar, 1997). As mentioned by Parthasarathy (1984), the summer monsoon rainfall is highly correlated

between some adjacent subdivisions in northwestern, central and peninsular India. It is therefore necessary to do empirical orthogonal function EOF analysis of the Indian summer monsoon rainfall to have a better idea about spatial distribution. Earlier Pant and Rupa Kumar (1997) has done EOF analysis on IITMS data to point out the spatial variability of Indian rainfall. On the basis of the rainfall data of India Meteorological Department south-west monsoon rainfall data (1875–1974) published by Banerjee and Raman (1976) and then by updating up to 1979, Rasmusson and Carpenter (1983) have done EOF analysis on 31 subdivisions (including Jammu and Kashmir and the hilly regions of north east India). It may be noted that at that time the country was represented by 31 meteorological subdivisions only. Using south-west monsoon season rainfall data of 36 subdivisions for the period 1901–2003 EOF analysis was carried out to address the spatial rainfall distribution. The most dominant EOF (Figure 10(a)) contributes 31% of the Indian monsoon rainfall while the second EOF explains EOF (Figure 10(b)) 16% of the total variance. The first EOF represents the dipole-type spatial structure in the variability of south-west monsoon rainfall. The north-eastern parts of India and Bihar region are in one phase while the rest of the country is in the opposite phase.

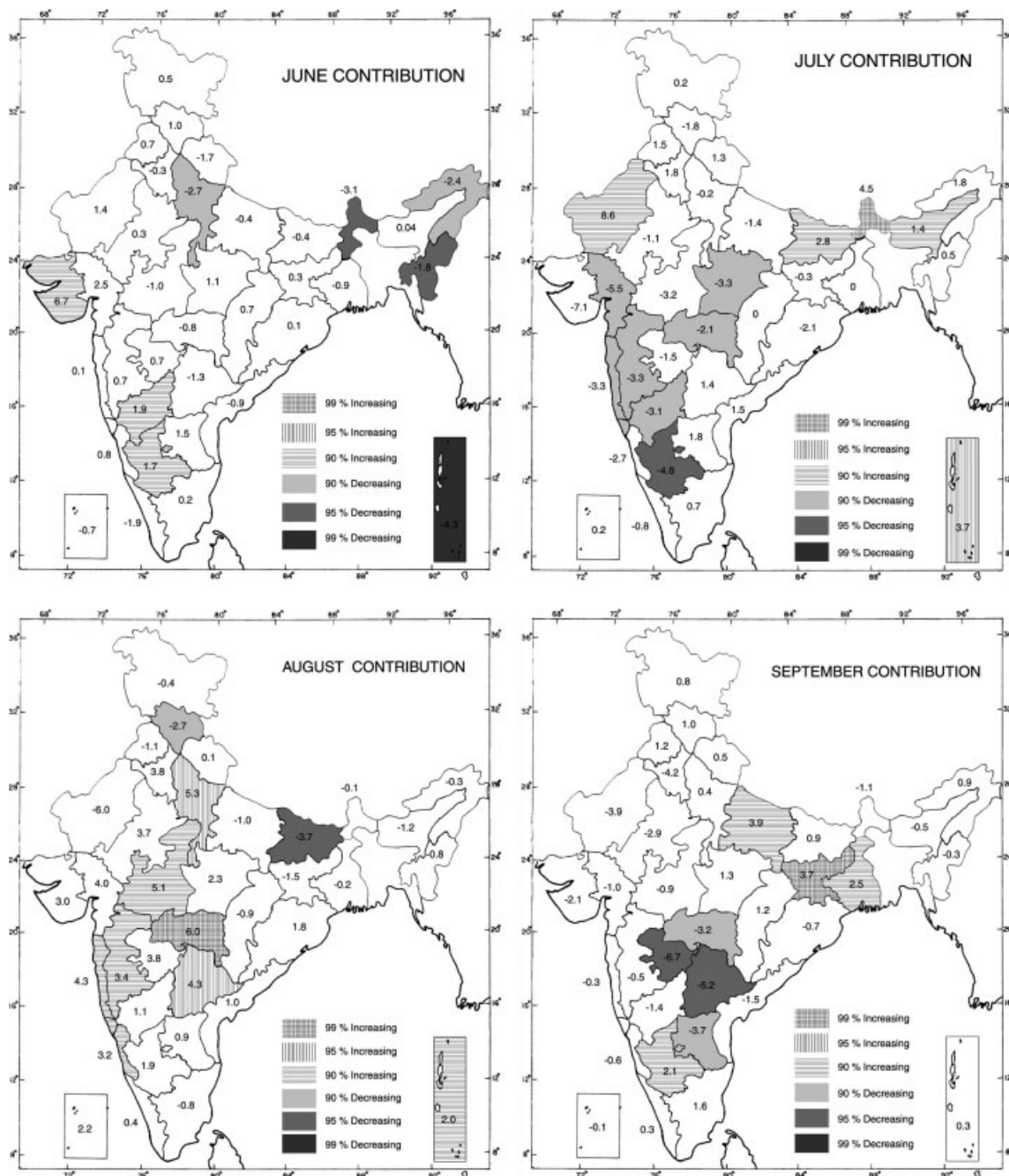


Figure 8. Increase/Decrease in rainfall in percentage in 100 years in monthly contribution of rainfall to annual rainfall for each of the four monsoon months for 36 subdivisions.

Interestingly, the dipole nature that exists in EOF 1 is not seen in the second EOF pattern, which shows a similar phase in all the regions. Third EOF (Figure 10(c)) accounts for 12% of the variance. The third EOF resembles similar pattern as that of spatial pattern of rainfall trends observed over different subdivisions (Figure 10). No significant trends are noticed in the first and second principal component time series. However the third principal component time series shows significant increasing trend. Both positive and negative loadings are present in third EOF with pockets of negative values over Kerala region and Chattisgarh region. To find out if there is any teleconnection of the south-west monsoon rainfall

with ENSO signals we have computed the correlation coefficient of south-west monsoon rainfall series and each of the first three principal component time series with the sea surface temperature SST anomalies over the Niño 3.4 region (5°N – 5°S , 120° – 170°W). The all India south-west monsoon rainfall is significantly and negatively correlated (Correlation coefficient: -0.57) with the SST anomalies over the Niño 3.4 region. The first EOF is significantly and positively correlated with the SST anomalies over the Niño 3.4 region (Correlation coefficient: 0.33). The second EOF is significantly and negatively correlated with the SST anomalies over the Niño 3.4 region (Correlation coefficient : 0.30). Third

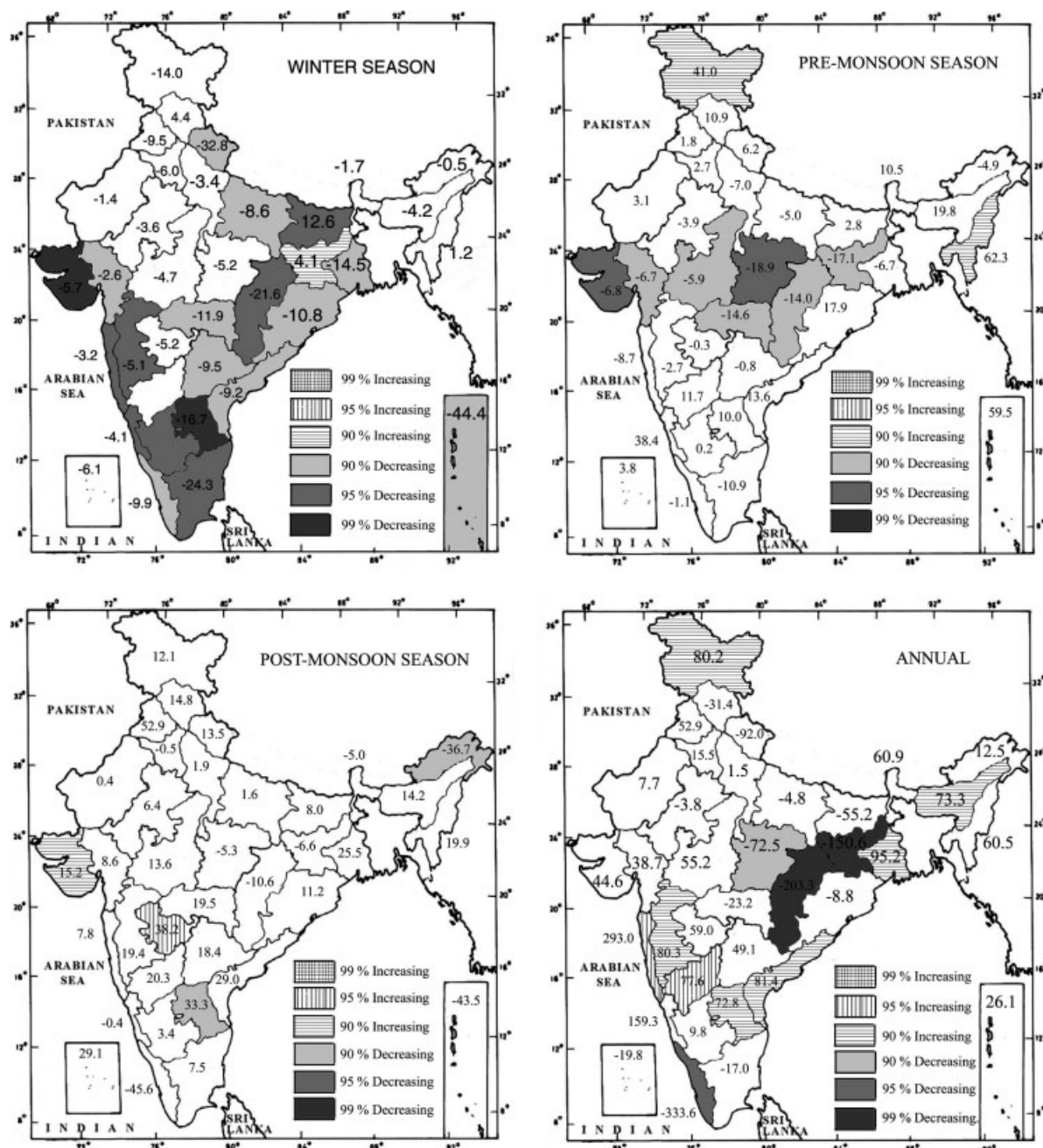


Figure 9. Increase/Decrease in rainfall in millimeter in 100 year in each of 36 subdivisions for the winter, pre-monsoon, post-monsoon seasons and annual. Different levels of significance are shaded.

EOF (Figure 10(c)), which accounts 12% of the variance is less correlated with the SST anomalies over the Niño 3.4 region (Correlation coefficient : -0.16). To see the stability of the relationship between the principal component time series of PC1, PC2 and PC 3 with SST anomalies of Niño 3.4 region, we have performed the 31 year running window correlation between these (Figure 11). Figure 11 also shows a 31-year running window correlation between all India seasonal monsoon rainfall and Niño 3.4 SST anomalies. The decrease in the strength of negative relationship of Niño 3.4 SST anomalies with all India south-west monsoon rainfall (AISMR) and positive relationship with PC1 from 1979 onwards can be noticed. However, there is not much change in the relationship of

Niño 3.4 SST anomalies with PC 2 and PC 3 during the period 1901–2003.

The main result regarding the change in rainfall pattern during south-west monsoon season can be summarized as increase in rainfall activities in Konkan and Goa and Maharashtra region and decrease in rainfall activities over Chattisgarh and adjacent areas. We have examined the changes in the circulation and synoptic patterns over the region, which may be useful to explain the corresponding changes in the rainfall pattern. The synoptic systems like low-pressure areas and depressions over Bay of Bengal are responsible for the rainfall activities over Chattisgarh, east Madhya Pradesh and adjacent areas as the low pressure systems forming over Bay of Bengal

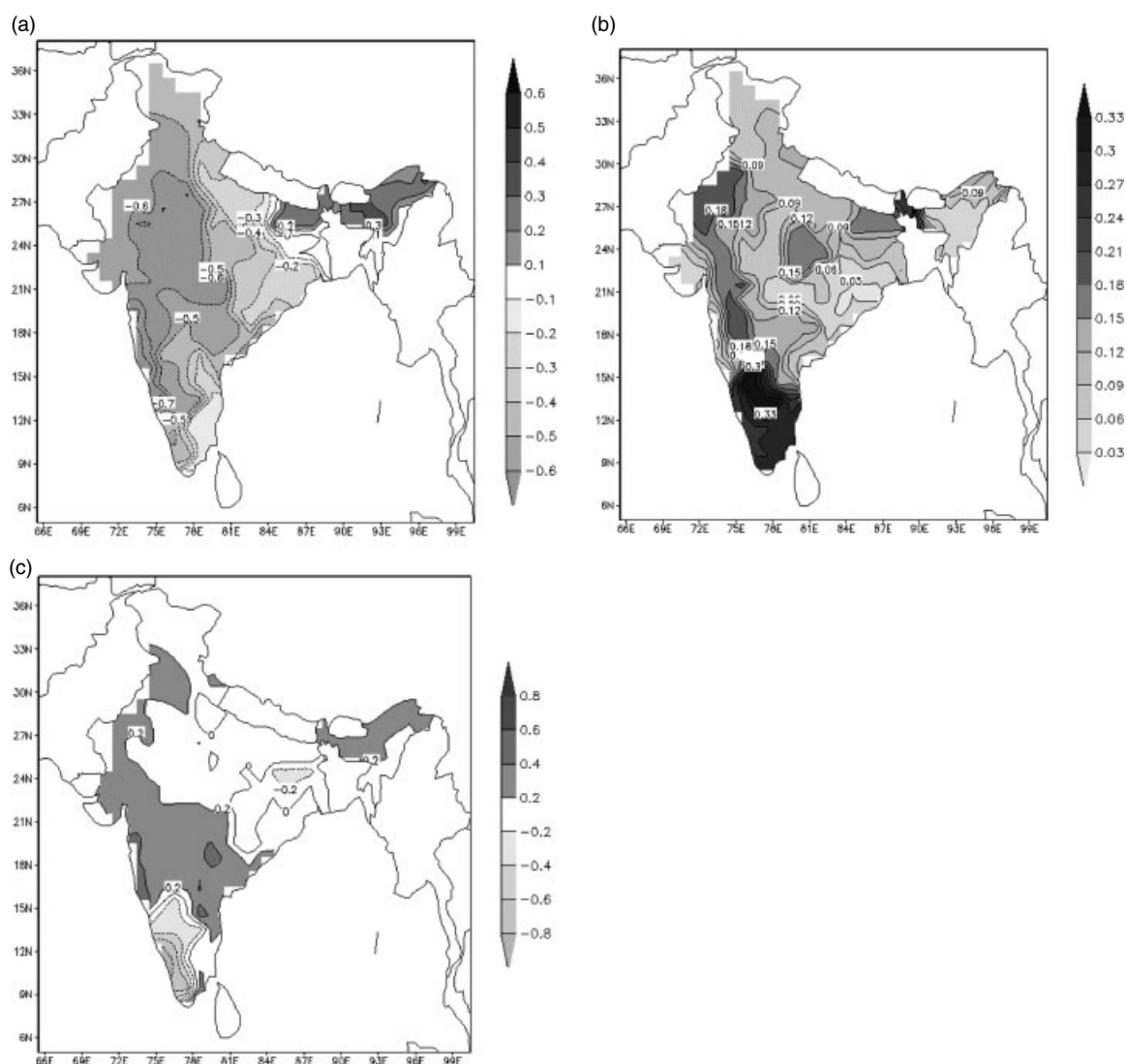


Figure 10. Spatial pattern of (a) first EOF which explains 31% of total variance. (b) Second EOF which explains 16% of total variance and (c) third EOF which explains 12% of total variance.

move in a northwesterly direction during the season. The activity of the monsoon trough and the formation of low pressure systems cause the increase in the formation of cloudiness over the region. Rajeevan *et al.* (2000) have studied the behaviour of SST and cloud cover properties over the Indian Ocean and Bay of Bengal for the period 1961–1998. The decrease in the activity of low cloudiness in spite of increase in SST over Bay of Bengal was noticed from 1977 onwards. Rajeevan *et al.* (2000) also indicated the decrease in the frequency of storm (depression and above) in the Bay of Bengal utilizing the data for the period 1901–1998. We have done the trend analysis on the frequency of depression and depression plus storms formed over the Bay of Bengal during the south-west monsoon season (June–September) (Figure 12). There is a significant decreasing trend in the frequency of both depressions and storms forming over north Bay of Bengal. Similar results were also obtained by Dash *et al.* (2004) by studying the data of different periods. Therefore, the decreasing trend in rainfall over

east central parts of India may be due to lesser frequency of low pressure systems forming over the Bay of Bengal. For establishing the physical causes in the increase in rainfall over the west coast and Maharashtra, more careful studies are required.

10. Conclusions

There was a need for development of a more homogeneous (spatially and temporally) rainfall time series for the Indian region using the latest data. The newly constructed rainfall series is uniformly distributed through out the country and it represents almost all the existing administrative districts. The present study brings out some of the interesting and also significant changes in the rainfall pattern of the country. The alternating sequence of multi-decadal periods of 30 years having frequent droughts and flood years are observed in the all India monsoon rainfall data. The decades 1961–1970,

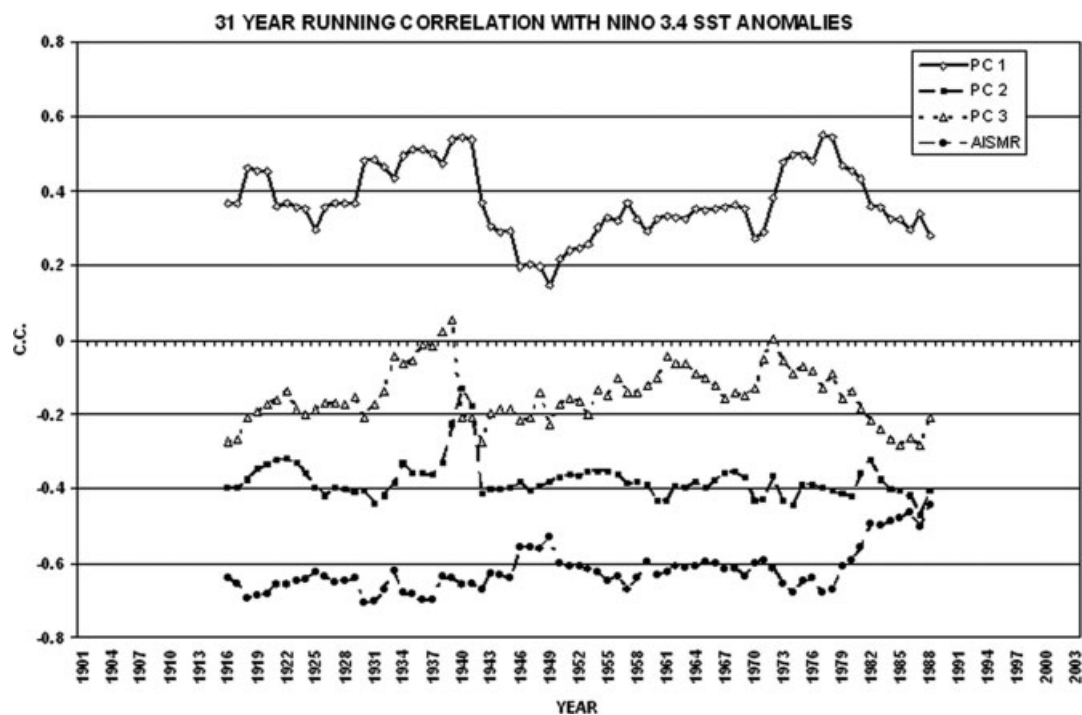


Figure 11. The 31 year running correlation between Niño 3.4 SST anomalies and first three principal component time series and all India south-west monsoon rainfall.

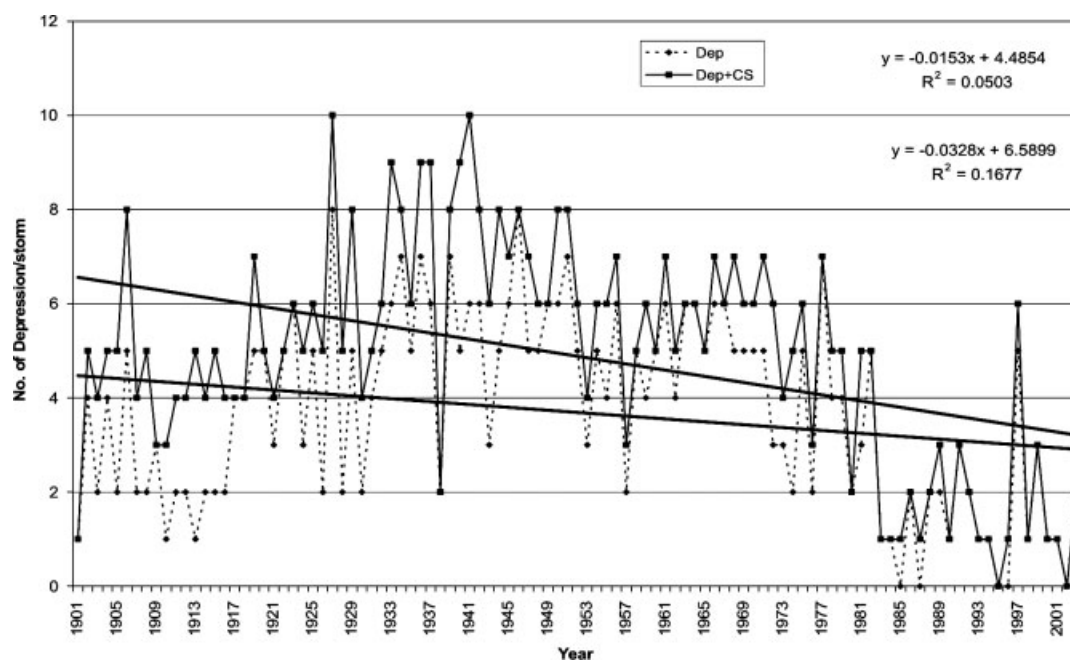


Figure 12. Trend in the frequency of Depression and Cyclonic storm during monsoon (JJAS) season over Bay of Bengal.

1971–1980 and 1981–1990 were dry periods. The first decade (1991–2000) in the next 30-year period already experienced wet period. Decadal variability is more for the June and September months while the decadal variability of July rainfall is almost similar to that of monsoon rainfall. The July rainfall has shown decreasing trends over most parts of central India. However, June and August rainfall has shown increasing trend over the

central and south western parts of the country. During the south-west monsoon season, three subdivisions viz. Jharkhand, Chattisgarh, Kerala showed significant decreasing trend and eight subdivisions viz. Gangetic WB, West UP, Jammu and Kashmir, Konkan and Goa, Madhya Maharashtra, Rayalseema, Coastal A P and North Interior Karnataka showed significant increasing trends. The study of trend or change in rainfall pattern of

other seasons reveals very interesting features and these are also supported by the changes in other meteorological features. The EOF analysis done on south-west monsoon rainfall separates EOF 1 with positive and significant correlation with Niño 3.4 SST anomalies while EOF 2 with negative and significant correlation with Niño 3.4 SST anomalies. The results brought out that there is weakening of the strength of negative relationship of Niño 3.4 SST anomalies with AISMR and positive relationship with PC1 from 1979 onwards.

Most of the studies on rainfall trend/analysis were only limited to seasonal monsoon rainfall only. For the first time, we have also studied contribution of each of the major rain producing month's (i.e. June, July, August and September) in annual rainfall and examine whether there is any significant change in their contribution. The June rainfall is getting importance as its contribution to annual rainfall is increasing in almost 20 subdivisions in which three are increasing significantly while decreasing in the remaining 16 subdivisions out of which three are significant. Contribution of July rainfall is decreasing in central and west peninsular India. But contribution of August rainfall is increasing in all these areas. Significant increasing trend is also observed in the annual rainfall for the subdivisions Konkan and Goa, Madhya Maharashtra, North Interior Karnataka, Rayalseema, coastal Andhra Pradesh, Gangetic West Bengal, Assam and Meghalaya and Jammu and Kashmir.

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References

- Banerjee AK, Raman CRV. 1976. One hundred years of southwest monsoon rainfall over India. Science Report No. 76/6. Meteorological Office: Pune; 7.
- Blanford H. 1886–1888. *Indian Meteorological Memoirs*, Viceroy and Governor General of India, Calcutta; Vol III: 638.
- Das MR, Mukhopadhyay RK, Dandekar MM, Kshirsagar SR. 2002. Pre-monsoon western disturbances in relation to monsoon rainfall, its advancement over NW India and their trends. *Current Science* **82**(11): 1320–1321.
- Dash SK, Kumar JR, Shekhar MS. 2004. On the decreasing frequency of monsoon depressions over the Indian region. *Current Science* **86**(10): 1404–1411.
- Jagannathan P, Bhalme HN. 1973. Changes in the pattern of distribution of southwest monsoon rainfall over India associated with sunspots. *Monthly Weather Review* **101**: 691–700.
- Jagannathan P, Parthasarathy B. 1973. Trends and periodicities of rainfall over India. *Monthly Weather Review* **101**: 371–375.
- Koteswaram P, Alvi SMA. 1969. Secular trends and periodicities in rainfall at west coast stations in India. *Current Science* **38**: 229–231.
- Mooley DA, Parthasarathy B. 1983. Variability of the Indian summer monsoon and tropical circulation features. *Monthly Weather Review* **111**: 967–968.
- Mooley DA, Parthasarathy B. 1984. Fluctuations in All-India summer monsoon rainfall during 1871–1978. *Climatic Change* **6**: 287–301.
- Normand C. 1953. Monsoon seasonal forecasting. *Quart J Roy. Meteor. Soc.* **79**: 463–473.
- Pant GB, Rupa Kumar K. 1997. *Climates of South Asia*. John Wiley & Sons: Chichester; 320.
- Parthasarathy B. 1984. Inter-annual and long term variability of Indian summer monsoon rainfall'. *Proceedings of the Indian Academy of Sciences (Earth Planetary Sciences)* **93**: 371–385.
- Parthasarathy B, Dhar ON. 1978. Climate fluctuations over Indian region—Rainfall: a review. Research Report no. RR-025. Indian Institute of Tropical Meteorology: Pune; 31.
- Parthasarathy B, Rupa Kumar K, Munot A. 1993. Homogeneous Indian monsoon rainfall: variability and prediction. *Proceedings of the Indian Academy of Sciences (Earth Planetary Sciences)* **102**: 121–155.
- Parthasarathy B, Munot A, Kothawale DR. 1994. All-India monthly and seasonal rainfall series 1887–1993. *Theoretical and Applied Climatology* **49**: 217–224.
- Pisharoty PR, Desai BN. 1956. Western disturbances and Indian weather. *Indian Journal of Meteorology Geophysics* **7**: 333–338.
- Pramanik SK, Jagannathan P. 1954. Climate change in India—I rainfall. *Indian Journal of Meteorology Geophysics* **4**: 291–309.
- Rajeevan M, De US, Prasad RK. 2000. Decadal variation of sea surface temperatures, cloudiness and monsoon depressions in the north Indian ocean. *Current Science* **79**(3): 283–285.
- Rasmusson EM, Carpenter TH. 1983. The Relationship between eastern equatorial pacific sea surface temperatures and rainfall over India and Srilanka. *Monthly Weather Review* **111**: 517–522.
- Rupa Kumar K, Pant GB, Parthasarathy B, Sontakke NA. 1992. Spatial and sub-seasonal patterns of the long-term trends of Indian summer monsoon rainfall. *International Journal of Climatology* **12**: 257–268.
- World Meteorological Organization. 1983. *Guide to Climatological Practices*, WMO No 100.